AVERAGE

LESSON-1: INTRODUCTION

$$\text{Average} = \frac{\text{Sum of Observation}}{\text{Total Number of Observation}}$$

Consecutive No:

$x, x+1, x+2, x+3, x+4$ ----

Consecutive Odd No:

$x, x+2, x+4, x+6, x+8$ ----

Consecutive even No:

$x, x+2, x+4, x+6, x+8$ ----

both the formula are same. Odd/even depends on value of x

* For some cases without using Average formula, By seeing the number itself we can tell its average But the number should fall any one of these category:

→ Consecutive No: $x, x+1, x+2, x+3,$ ----

Eg: $x = 90$

90, 91, 92, 93, 94

If it is a consecutive number, then the centre number will be its average

90, 91, 92, 93, 94

average = 92

Eg: 101, 102, 103, 104

here centre number will be 102.5

so average = 102.5

→ Consecutive Odd No: $x, x+2, x+4,$ ----

Eg: $x = 1$ (odd)

1, 3, 5, 7, 9

1, 3, 5, 7, 9

If it is a consecutive odd number, then the centre number will be it's average

average = 5

→ Consecutive even number: $x, x+2, x+4, x+6$ ----

Eg: $x = 26$

26, 28, 30, 32, 34

If it is a consecutive even number, then the centre number will be it's average

average = 30

Eg: 52, 54, 56, 58

It is a consecutive even number

average = centre No. = 55

LESSON - 2 : BASIC QUESTION

Q) The average age of A, B and C is 26 years, if the average age of A and C is 29 years, what is the age of B in Years?

Sol: Average age of A, B, C = 26

i.e. $\frac{A+B+C}{3} = 26$

$A+B+C = 78$ —— ①

Average age of A, C = 29

$\frac{A+C}{2} = 29$

$A+C = 58$ —— ②

age of B = ?

Put ② in ①

$A + B + C = 78$

$B + 58 = 78 \implies B = 78 - 58 = 20$

$\therefore B = 20$ years.

9) The average of 7 numbers is 5. If the average of first six of these numbers is 4, the seventh number is?

Sol: Average of 7 numbers $= 5$

i.e. $\frac{\text{Total sum}}{7} = 5$

Total sum $= 35$ —— ①

Average of 1st 6 numbers $= 4$

i.e. $\frac{n_1 + n_2 + \cdots n_6}{6} = 4$

$n_1 + n_2 + \cdots n_6 = 24$ —— ②

From ① $\Rightarrow n_1 + n_2 + n_3 + n_4 + n_5 + n_6 + n_7 = 35$

put ②

$[n_1 + n_2 + \cdots + n_6] + n_7 = 35$

$24 + n_7 = 35$

$n_7 = 35 - 24$

$n_7 = 11$

9) The average of number 10 number is 7. What will be the new average if each of the number is multiplied by 8?

Sol: Average of 10 numbers $= 7$

i.e. $\frac{\text{Total sum}}{10} = 7$

Total sum $= 70$

$n_1 + \cdots n_{10} = 70$

each number is multiplied by 8

i.e. $8 \times [n_1 + \cdots n_{10}] = \cancel{8 \times 70}$

8×70

Total sum of new numbers =

$N_1 + --- N_{10} = 560$

Average of new numbers $= \frac{N_1 + --- N_{10}}{10} = \frac{56\cancel{0}}{1\cancel{0}}$

$= 56$

Q) The average of five consecutive even numbers starting with 4, is

Sol: five consecutive even numbers $= x, x+2, x+4, x+6, x+8$

$x = 4$

$= 4, 6, \underline{8}, 10, 12$

average $= 8$

Q) A, B, C and D are four consecutive even numbers, respectively and their average is 65, what the product of A and D.

Sol: $\underset{x}{A} \quad \underset{x+2}{B} \quad \underset{65}{\downarrow} \quad \underset{x+4}{C} \quad \underset{x+6}{D} \Rightarrow 62 \quad 64 \quad \underset{65}{\downarrow} \quad 66 \quad 68$

$A \times D = 62 \times 68 = 4216$

Q) A, B, C and D are four consecutive odd numbers and their average is 42. what is the product of B and D?

Sol: $\underset{x}{A} \quad \underset{x+2}{B} \quad \underset{42}{\downarrow} \quad \underset{x+4}{C} \quad \underset{x+6}{D} \Rightarrow 39 \quad 41 \quad \underset{42}{\downarrow} \quad 43 \quad 45$

$B \times D = 41 \times 45$

$= 1845$

9) Of the three numbers. The first is twice the second and the second is thrice the third. If the average of the three numbers is 10. The numbers are:

Sol: Let the 3 numbers be A, B, C

$A = 2B$ $\qquad \frac{A+B+C}{3} = 10$

$B = 3C$

$A + B + C = 30$

$2B + B + C = 30$

$2[3C] + 3C + C = 30$

$6C + 4C = 30$

$10C = 30$

$\boxed{C = 3}$

$B = 3C = 3(3) \Rightarrow \boxed{B = 9}$

$A = 2B = 2(9)$

$\boxed{A = 18}$

9) The sum of five numbers is 555. The average of the first two numbers is 75 and the third number is 115. What is the average of the last two numbers?

Sol: $n_1 + n_2 + n_3 + n_4 + n_5 = 555$

$\frac{n_1 + n_2}{2} = 75 \Rightarrow n_1 + n_2 = 75 \times 2 = 150$

$n_3 = 115$

$[n_1 + n_2] + n_3 + n_4 + n_5 = 555$

$150 + 115 + n_4 + n_5 = 555$

$n_4 + n_5 = 555 - (150 + 115)$

$= 555 - 265$

$= 290$

average $= \frac{n_4 + n_5}{2} = \frac{290}{2} = 145$

Q) The average expenditure of a man for the first five months is Rs. 3600 and for next seven months it is Rs. 3900. If he saves Rs: 8700 during the year, his average income per month is.

Sol: Average $= \frac{\text{Total}}{n}$

Total expenditure of 1^{st} 5 monts $= 5 \times 3600$
$= 18{,}000$

Total expenditure of next 7 months $= 7 \times 3900$
$= 27300$

Total expenditure for 12 months $= 18000 + 27300$
$= 45300$

Total savings $= 8700$

Income $=$ saving $+$ Exp
$= 8700 + 45300$
$= 54000$

Average income $= \frac{54000}{12} = 4500$

Q) The sum of three numbers is 98. If the ratio between first and second be 2:3 and between second and third be 5:8, then the second number is;

Sol: $n_1 + n_2 + n_3 = 98$

$\frac{n_1}{n_2} = \frac{2}{3}$ $\qquad$ $\frac{n_2}{n_3} = \frac{5}{8}$

$n_1 = \frac{2n_2}{3}$ $\qquad$ $n_3 = \frac{8}{5} n_2$

$n_1 + n_2 + n_3 = 98$

$\frac{2}{3} n_2 + n_2 + \frac{8}{5} n_2 = 98$

$$n_2\left[\frac{10+15+24}{15}\right] = 98$$

$$n_2 = \frac{\cancel{98}^{2} \times 15}{\cancel{49}_{1}} = 30$$

Method 2 : Using ratio and proportion

$a+b+c = 98$

$a:b = 2:3 \qquad b:c = 5:8$

$$\begin{array}{rcl} a:b:c & = & 2:3 \\ & & \quad\; 5:8 \\ \hline & & 10:15:24 \end{array}$$

$\because a:b:c = 10:15:24.$

$a = 10x, \; b = 15x, \; c = 24x$

$a+b+c = 10x + 15x + 24x = 98$

$49x = 98$

$x = 2$

$\therefore \; b = 15x = 15(2) = 30$

LESSON-3 : Based on Equation

Q) The average of marks obtained by 120 candidates was 35. If the average of marks of passed candidates was 39 and that of failed candidates was 15, the number of candidates who passed the examination is :

Let total no. candidates = N

passed candidates = n_p

failed candidates = n_f

$$n_p + n_f = N$$

$\therefore n_p + n_f = 120$ $[\because N = 120]$

Let $n_p = x$

$n_f = 120 - x$

Average of 120 candidates = 35

i.e. $\dfrac{\text{Total sum}}{120} = 35$

Total sum = 120×35 = 4200

Average of passed candidates = 39

i.e. $\dfrac{\text{sum of passed}}{n_p} = 39$

Sum of passed = $39 n_p$

$= 39(x)$

Average of failed candidates = 15

i.e. $\dfrac{\text{sum of failed}}{n_f} = 15$

Sum of failed = $15(n_f)$

$= 15(120 - x)$

Total sum = Sum of passed + Sum of failed

$4200 = 39[x] + 15[120 - x]$

$4200 = 39x + 1800 - 15x$

$4200 = 24x + 1800$

$24x = 4200 - 1800$

$24x = 2400$

$x = 100$

$\therefore$ no. of passed $= n_p = x = 100$

9) In a school, the average age of students is 6 years and the average age of 12 teachers is ~~yo~~ 40 years. If the average age of the combined group of all the teachers and the students is 7 years, then the number of students is:

Sol: No. of teachers = 12

No. of students = x (let)

Average age of teachers & students = 7

i.e. $\dfrac{\text{Total sum}}{12 + x} = 7$

Total sum = $7[12 + x]$

Average age of students = 6

i.e. $\dfrac{\text{sum of students}}{x} = 6$

Sum of students = $6x$

Average age of 12 teachers = 40

i.e. $\dfrac{\text{sum of teachers}}{12} = 40$

Sum of teachers = 12×40

= 480

Total sum = sum of students + sum of teachers

$7[12+x] = 6x + 480$

$84 + 7x = 6x + 480$

$x = 480 - 84$

$x = 396$

8) The average monthly salary of all the employees in an industry is Rs. 12000. The average salary of male employees is Rs. 15000 and that of female employees is Rs. 8000. What is the ratio of male employees to female employees?

Sol: Average of all employees = 12000

i.e. $\dfrac{\text{Total sum}}{N} = 12000$

Total sum = $12000(N)$

Let no. of male = x

no. of female = y

$$N = x + y$$

$\therefore$ Total sum = $12000[x+y]$

Average of male = 15000

i.e. $\frac{\text{sum of male}}{x} = 15000$

Sum of male = $15000x$

Average of female = 8000

i.e. $\frac{\text{sum of female}}{y} = 8000$

Sum of female = $8000y$

Total sum = sum of male + sum of female

$$12000[x+y] = 15000x + 8000y$$

$$12000y - 8000y = 15000x - 12000x$$

$$4\cancel{000}y = 3\cancel{000}x$$

$$\frac{x}{y} = \frac{4}{3}$$

9) In a school with 600 students, the average age of the boys is 12 years and that of the girls is 11 years. If the average age of the school is 11 years and 9 months, then the number of girls in the school is:

Sol: $N = 600$

600 → boys x, girls $600-x$

Average of boys = 12 years

$= 12 \times 12$ months

$= 144$ months

i.e. $\frac{\text{sum of boys}}{x} = 144$

Sum of boys = $144x$

Average of girls = 11 years

$= 11 \times 12 = 132$ months

i.e. $\frac{\text{sum of girls}}{600-x} = 132$

Sum of girls = $132(600-x)$

Average of school = 11 years and 9 months

$= [11 \times 12 \text{ months}] + 9 \text{ months}$

$= 132 + 9$

$= 141$ months

i.e. $\frac{\text{Total sum}}{600} = 141$

Total sum $= 600 \times 141$

$= 84600$

Total Sum = Sum of boys + sum of girls

$84600 = 144x + 132(600 - x)$

$84600 = 144x + 79200 - 132x$

$84600 - 79200 = 12x$

$5400 = 12x$

$x = 450$

No. of girls $= 600 - x$

$= 600 - 450$

$= 150$

LESSON - 4 : TRUE/FALSE AVERAGE

Q) The mean of the marks obtained by 100 students is 60. If the marks obtained by one of the students was incorrectly calculated as 75, whereas the actual marks obtained by him was 65, what is the correct mean of the marks obtained by the students?

Sol: * Mean and Average both are same

Average of 100 students = 60

i.e. $\frac{\text{Total Sum}}{100} = 60$

Total Sum = 6000

For one Student

Actual mark = 65 $\xrightarrow{+10}$ wrong mark = 75

So subtract 10 from total sum & find new avg.

$$\text{Correct average} = \frac{6000-10}{100} = \frac{5990}{100}$$

$$= 59.9$$

8) A Mathematics teacher tabulated the marks secured by 35 students of 8th class. The average of their marks was 72. If the marks secured by Reema was written as 36 instead of 86 then find the correct average marks up to two decimal places.

Sol: No. of students = 35

Calculated average = 72

i.e. $\frac{\text{Total sum}}{35} = 72$

Total sum = 35×72 = 2520

wrong mark = 36 $\xrightarrow{+50}$ original mark = 86

So, Add 50 to sum

$$\text{Correct average} = \frac{2520+50}{35} = \frac{\cancel{2570}^{514}}{\cancel{35}_{7}} \quad 73.43$$

$$= 73.43$$

Q) The average of marks of 14 students was calculated as 71. But, it was later found that the marks of one student had been wrongly entered as 42 instead of 56 and of another as 74 instead of 32. The correct average is.

Sol: Average of 14 students = 71

i.e. $\frac{\text{Total sum}}{14} = 71$

Calculated total sum $= 14 \times 71$

$= 994$

student1: taken wrong mark = 42 $\xrightarrow{+14}$ correct mark = 56

student2: taken wrong mark = 74 $\xrightarrow{-42}$ Correct mark = 32

New total sum $= 994 + 14 - 42$

$= 966$

New average $= \frac{\cancel{966}^{\cancel{483}\,69}}{\cancel{14}_{\cancel{7}}} = 69$

Q) The average marks in science subject of a class of 20 students is 68. If the marks of two students were misread as 48 and 65 of the actual marks 72 and 61, respectively, then what would be the correct average?

Sol: N = 20

Average of 20 students = 68

i.e. $\frac{\text{Total sum}}{20} = 68$

Total Sum $= 68 \times 20$

$= 1360$

wrong marks = 48 $\xrightarrow{+24}$ correct marks = 72

= 65 $\xrightarrow{-4}$ = 61

New Total Sum = 1360 + 24 - 4
= 1380

Correct average = $\frac{\cancel{1380}^{69}}{\cancel{20}}$ = 69

LESSON-5 : Replacing a Person.

Q) The average age of a committee of 8 members is 40 years. A member, aged 55 years, retired and he was replaced by a member aged 39 years. The average age of the present committee is:

Sol: Total average of 8 members = 40

i.e. $\frac{\text{Total Sum}}{8}$ = 40

∴ Total sum = 320

retired person = 55   replaced person = 39

i.e. New Total sum = 320 - 55 + 39
= 304

New average = $\frac{\cancel{304}^{38}}{\cancel{8}}$ = 38

Q) The average weight of 3 men A, B and C is 84 kg. another man, D, joins the group, and the average weight becomes 80 kg. If another man, E, whose weight 3 kg more than that of D, replaces A, then average weight of B, C, D and E becomes 79 kg. The weight of A is:

Sol: Average of A, B, C = 84

i.e. $\frac{A+B+C}{3} = 84$

$A+B+C = 252$

D Joins:

Average of A, B, C, D = 80

i.e. $A+B+C+D = 320$

weight of D = $[\cancel{A}+\cancel{B}+\cancel{C}+D] - [\cancel{A}+\cancel{B}+\cancel{C}] = 320 - 252$

$D = 68$ kg

E:

$E = 3+D = 3+68$

$E = 71$ kg

A removed & E added:

New sum = $B+C+D+E$

New Average = $\frac{B+C+D+E}{4} = 79$

$B+C+D+E = 316$

$B+C+68+71 = 316$

$B+C = 316-139$

$\boxed{B+C = 177}$

$\because$ $A+B+C = 252$

$A = 252 - 177$

$\boxed{A = 75 \text{ kg}}$

LESSON - 6 : INCLUDING / EXCLUDING

Q) The average weight of 21 boys was recorded as 64 kg. If the weight of the teacher was added the average increased by one kg. what was the teachers age?

Sol: average of 21 boys = 64

Total sum = 64×21 = 1344

Teacher added:

new average of 22 = 65

Total sum = 65×22 = 1430

Teachers age = 1430 − 1344

= ~~86~~ 86

Q) The average age of 14 girls and their teacher's age is 15 yr. If the teacher's age is excluded then the average reduced by 1. what is the teachers age?

Sol: Average of 14 girls and teacher = 15

Total sum = 15×15 = 225

Teacher removed:

New average of 14 = 14

Total sum = 14×14 = 196

Teacher's age = 225 − 196

= 29 yr

Q) The average age of 5 members of a family is 25 yr. If the servant of the family is included the average age increased by 40%. what is the age of the servant?

Sol: Average of 5 = 25

Total sum = 5×25 = 125

Servant added:

new average increased by 40%

100% $\longrightarrow$ 25 avg
140% $\longrightarrow$? $\Bigg\}$ $= \dfrac{\cancel{25} \times \cancel{140}}{2\cancel{100}} = 35$

$\therefore$ New average6 = 35

New Tobal sum = $6 \times 35 = 210$

Servant age = $210 - 125$
= 85 yr

8) The average age of the class is 35 yr. 6 new students with an average age of 33 yr joined in that class, there by decreasing the average by half year. The Original strength of the class was?

Sol: Let initially no. of people in class $= x$

Average age of x people in class = 35

Total sum of x people $= 35x$

6 new students added:

Average of these 6 students = 33

Sum of these 6 people $= 33 \times 6$
$= 198$

~~Total new ave~~

~~Total new average = 35 + 33 = 68~~

Total new sum $= 35x + 198$

Total new average $= \dfrac{35x + 198}{x+6} = 35 - \dfrac{1}{2}$

$2(35x + 198) = (x+6)(70-1)$

$70x + 396 = 69x + 414$

$x = 414 - 396$

$x = 018$

$\therefore \boxed{x = 18}$

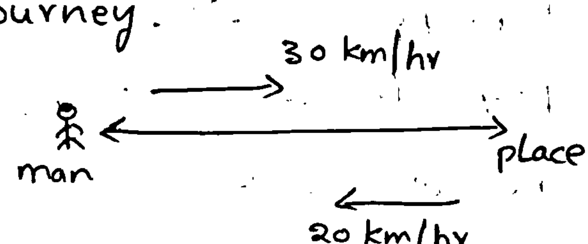
LESSON 7: AVERAGE SPEED

* If the certain distance is covered at the speed of x km/hr and the same distance is covered at y km/hr, then the average speed during entire journey $= \left(\frac{2xy}{x+y}\right)$ km/hr

here, distance is same in both the cases

Q) A man goes to a certain place at a speed of 30 km/hr and returns to original place at a speed of 20 km/hr. Find out the average speed during the entire journey.

Sol:

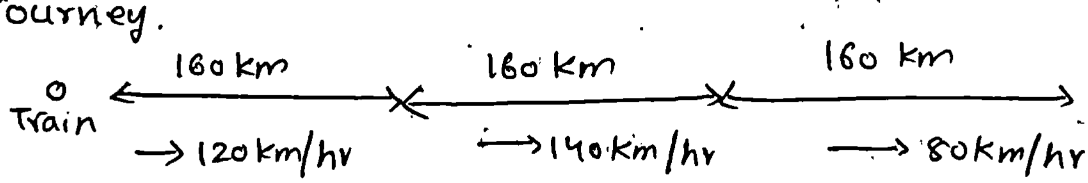


$$\text{average speed} = \frac{2(30)(20)}{30+20} = \frac{2(\overset{6}{\cancel{30}})(2\cancel{0})}{\cancel{5}\cancel{0}}$$

$$= 24 \text{ km/hr}$$

Q) A train covers the first 160 km at a speed of 120 km/hr, another 160 km at 140 km/hr and last 160 km at 80 km/hr. Find out the average speed of the train for entire journey.

Sol:



In all 3 cases, distance travelled is same

$$\therefore \text{average speed in entire journey} = \frac{3xyz}{xy+yz+zx}$$

$$= \frac{3(120)(140)(80)}{(120)(140) + (140)(80) + (80)(120)}$$

9) A person covers 9km at a speed of 3 Km/hr, 25 km at a speed of 5 Km/hr and 30 km at a speed of 10 Km/hr. Find out the average speed of the entire journey?

Sol: * If the person covers A km at a speed of x km/hr, B km at a speed of y km/hr and C km at a speed of z km/hr. Find out average speed of entire journey.

$$\left(\frac{A + B + C}{\frac{A}{x} + \frac{B}{y} + \frac{C}{z}} \right) \text{ km/hr}$$

$$\frac{9 + 25 + 30}{\frac{\cancel{9}^{3}}{\cancel{3}} + \frac{\cancel{25}^{5}}{\cancel{5}} + \frac{3\cancel{0}}{1\cancel{0}}} = \frac{9 + 25 + 30}{3 + 5 + 3} = \frac{44}{11} = 4 \text{ km/h}$$